

1 Article

2 **Prediction-Driven Non-Preemptive Scheduling with Bounded**  
3 **Overtaking for Shared Execution Services under**  
4 **Deadline-Driven Bursty Load**5 Yazhou Guo <sup>1,\*</sup>, Zexin Lin <sup>2</sup>, Chengyang Huo <sup>3</sup> and Yurong Song <sup>1</sup>6 <sup>1</sup> College of Automation, Nanjing University of Posts and Telecommunications, Nanjing 210023, China;  
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13 **Abstract**

Services that grade students' programs, run cloud functions or test software run many users' jobs on  $k$  shared servers without interruption, each under a time limit  $L$ , with bursty arrivals. Ordering the queue by predicted running time shortens most waits, but one wrong prediction can hold a job back for a long time. What can such a service promise every job when predictions may be arbitrarily wrong? Under any rule that never idles a server while jobs wait and never interrupts a job, a job's extra wait relative to first-come first-served (FCFS) equals the net running time of the jobs that overtook it, divided by  $k$ , up to an error of  $(2 - 2/k)L$  that cannot be improved. Limiting that running time is therefore sufficient for a per-job promise and, up to an additive  $O(kL)$ , necessary. A guard enforcing such a limit bounds every job's wait against FCFS for any arrivals, predictions and underlying policy, with exact constant  $(3 - 2/k)L$ ; if completion reports arrive up to  $\delta$  late, the bound grows by  $L + \delta$ . On a university grading log at peak load, scoring each job only from results available at its arrival, a guard promising 600 seconds keeps 0.705 of the reduction in 99th-percentile wait that shortest-job-first achieves over FCFS, and 0.894 on three terms held back until the method was frozen.

**Keywords:** non-preemptive scheduling; scheduling with predictions; multiserver queues; bounded overtaking; per-job guarantee; first-come first-served; runtime prediction; cluster scheduling; heterogeneous graph neural network

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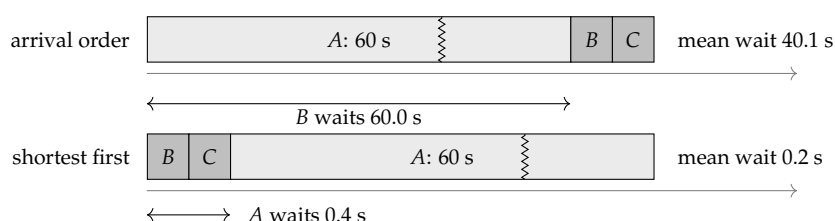
33 **1. Introduction**

Many computing services let a large number of users share a fixed set of machines. A university system that compiles and tests every programming assignment its students submit works this way [1]; so does a service that runs a software project's tests each time a developer changes the code, known as continuous integration (CI), and a cloud platform that runs small user functions on request. Each submission becomes a job, and the jobs wait in one queue for  $k$  identical machines, which we call servers. A server that becomes free takes the next job and runs it without interruption, until the job finishes or reaches a

time limit  $L$  that the service imposes on every job, and then reports the completion to the scheduler over the network.

Two features of the demand make such a service slowest exactly when it is busiest. Submissions follow external calendars, such as a coursework deadline or a release date, so they arrive in bursts, and the crowding before a fixed cutoff has been measured directly [2–4]. A small fraction of the jobs also carries most of the work. In the log of cloud-function calls we use [5,6], the longest 1% of the calls account for 44% of the total running time. In the two logs of students' programming submissions, with each running time capped at that platform's time limit, the longest 1% account for 37.5% and 27.5%. The university grading system shows both features at their sharpest, and it is our first case. Before a common deadline, submissions that looped until their time limit occupied the graders, and the waits of everything queued behind them grew in a cascade [7]. The manual of a programming-contest judging system recommends one judging machine per 20 teams and warns that waits grow with fewer machines [8].

Ordering the queue by predicted cost exploits the second property, as a queue of three jobs on one server shows (Figure 1). Job  $A$  loops and is stopped by the 60-second limit; jobs  $B$  and  $C$  take 0.2 seconds each. In arrival order the mean wait is 40.1 seconds; shortest first, it is 0.2 seconds, at a cost of 0.4 seconds to  $A$ . These numbers are arithmetic and rest on one assumption, that the scheduler knows before  $A$  starts that  $A$  will be long. When an estimate is wrong, the same ordering can hold a job back for a long time. In our simulation of the busiest period we tested, driven by the recorded submissions, one job waited almost a hundred time limits longer than it would have under first-come first-served (FCFS) order (Section 8.4).



**Figure 1.** The three-job example of Section 1 on one server:  $A$  is stopped at the per-job limit  $L = 60$  s,  $B$  and  $C$  take 0.2 s each, and all three are present at  $t = 0$ . The boxes are schematic, not to scale, and the zigzag marks the elided part of  $A$ . The two orders execute the same total work; reordering moves 60 seconds of waiting off  $B$  and  $C$  and puts 0.4 seconds onto  $A$ .

A line of work on algorithms with predictions bounds how an algorithm performs as a function of the error of the predictions it is given [9,10]. For scheduling, these bounds compare the total or mean completion time with that of an optimal schedule, and the constant-factor results assume that the prediction error is bounded [11–15]. A predictor in service carries no such bound, and an operator is asked about individual jobs rather than averages. Two results come closer: one bounds the worst slowdown of any job on a machine that may interrupt jobs [16], and one bounds a request's delay in a queue with steady, light-tailed demand [17,18]. We have not found a guarantee that is stated for every job against FCFS, holds for any arrival sequence and any predictions, allows no interruption, covers  $k$  servers, and can be enforced while any scheduling rule runs underneath. This paper answers three questions.

1. Can a job's running time be estimated when it arrives, from the history available at that moment?
2. How should a scheduler that cannot interrupt jobs order the queue once it has such estimates?

3. What can the service promise every job when the estimates are wrong, and when the completion reports it relies on arrive late?

For the third question we change the reference point instead of assuming more about the predictions. For each job, we bound how much longer it waits than it would have under FCFS on the same sequence of submissions, with no assumption about the arrival process or about prediction quality. The mechanism is an overtake budget. For every waiting job the scheduler adds up the running time of the jobs that arrived later but were started ahead of it, and as soon as that total reaches the job's budget, the job is served before any later arrival. This guard sits on top of any scheduling rule, including one driven by a learned predictor, and changes only which job starts next; a cap  $B_{\max}$  on the budget fixes the promise. The budget counts running time rather than the number of jobs that overtake, because of an identity we prove: under any scheduling rule that never leaves a server idle while jobs wait and never interrupts a job, a job's extra wait relative to FCFS equals the net running time of the jobs that overtook it, divided by  $k$ , up to an error of at most  $(2 - 2/k)L$ . The guard learns of completions from the servers' reports, so we also bound what it promises when these reports cross a network and arrive late.

For the first two questions we compare three kinds of predictor, trained and tested on the same data under the same rule about what may be seen when a job arrives: gradient-boosted decision trees [19], a gated recurrent unit (GRU) that reads each user's earlier jobs, and a heterogeneous graph neural network (GNN) over users, programming tasks and assignments, of the relational graph convolutional network (R-GCN) type [20]. The history available at arrival separates the long jobs from the short ones on all three platforms we test it on. The graph network does not improve on the trees on average, and its edges add only 0.006 in Spearman correlation over all test submissions, so most of the information lies in each user's and each task's own history. On tasks with fewer than five earlier results the graph network does rank jobs better, by 0.146 in Spearman correlation over the trees and by 0.028 over the same network with shuffled edges (Section 4). The scheduler orders jobs by their expected running time given the observed features, and for our fitted predictor, treating a long job as short costs about ten times more than treating a short job as long (Section 8.3).

This study makes three contributions.

1. We prove the identity above and show that its error constant  $(2 - 2/k)L$  cannot be improved. Conversely, any per-job promise of this kind limits the running time of the jobs that overtake a job, up to an additive  $O(kL)$ , whereas a limit on how many jobs overtake it is not required and, on our main log, is looser by a factor of 150.
2. We give the guard and prove that it keeps every job's extra wait over FCFS below  $B_{\max}/k + (3 - 2/k)L$ , for  $k$  servers, any arrivals and any predictions; the constant cannot be improved. An operator who wants to promise  $G$  seconds sets  $B_{\max} = k(G - (3 - 2/k)L)$  and needs no data to do so. When completion reports arrive up to  $\delta$  late, the promise grows by  $L + \delta$ , and it does not grow at all if each server reports its own completion before taking its next job. We also bound the absolute wait from the arrival record alone, and compare the guard with a maximum waiting time, which makes a promise of the same form.
3. We evaluate the method on six logs: two from university systems that grade students' programming submissions [21,22], one from a cloud-function platform [5], two from CI machine pools [23,24] and one from an industrial compute farm [25,26]. Every simulated job is checked against the proved bound. At the busiest load, with every score computed from the results available when its job arrives, a guard that promises 600 seconds keeps 0.705 (bootstrap interval [0.648, 0.768]) of the reduction in the 99th-

percentile wait before deadlines that shortest job first (SJF), which knows every running time in advance, achieves over FCFS (Section 8.1).

Two qualifications apply throughout. The grading logs record how long each submission ran but not how long it waited, so the shared queue in our simulations is our own construction (Section 7). Most numbers reported come from data we used while developing the method. Three academic terms of the main log were held back unread and opened once, after the method and its parameters were frozen. On them the guard keeps 0.894 of the reduction at peak load, against 0.705 on the development terms (Section 8.2).

The rest of the paper is organised as follows. Section 2 reviews related work, and Section 3 states the model and the measures of performance. Section 4 compares the predictors, Section 5 describes the scheduler and the guard, and Section 6 proves the identity and the per-job bounds. Section 7 describes the logs, Section 8 reports the experiments and Section 9 discusses the limitations; longer proofs are in the Supplementary Materials.

## 2. Related Work

### 2.1. Scheduling with Predictions

Algorithms with predictions are judged by consistency, how well they do when the predictions are exact, and robustness, how well they do when the predictions are arbitrarily wrong [9,10]. In scheduling, predicted sizes have been used to minimise the total completion time of jobs whose sizes are unknown [11], learned machine weights to assign jobs online [12], and a predicted order of the jobs in place of their sizes [13]. Queueing models treat the prediction as a random variable [27], bound its error multiplicatively [14] or measure it as a distortion [28,29], and further work covers many-server limits [30], the cost of obtaining predictions [31] and the worst slowdown of any job on a machine that may interrupt jobs [16]; Mitzenmacher and Shahout survey this line [15]. These results bound a mean or a competitive ratio against an optimal schedule, and the bound degrades with a measure of the prediction error. In their terms, our guarantee is a robustness statement made for each job and on every sample path: it holds for any predictions, but it bounds the loss against FCFS and does not by itself promise a gain, which Section 8 measures.

### 2.2. Per-Job Guarantees and Bounded Overtaking

Schwiegelshohn and Yahyapour call a schedule  $\lambda$ -fair when later submissions cannot increase a job's flow time by more than a factor  $\lambda$ , and bound fairness and efficiency together for a policy that interrupts whole groups of processors [32]. Batch schedulers measure a job's delay against a fair start time, obtained by simulating the schedule again without the later arrivals [33,34]. Conservative backfilling gives every queued job a reservation that later jobs cannot delay [35,36], slack-based backfilling lets later jobs delay a job only within a slack fixed at submission [37], and a preemptive variant of the Extensible Argonne Scheduling sYstem (EASY) backfilling [38] kills and restarts the backfilled jobs that would delay a higher-priority one [39]. These guarantees rest on reservations or on interrupting jobs; ours is additive, needs no interruption and is measured against FCFS on the same input.

Bounded overtaking has been studied in two units. The first is a count: Nudge lets a small arriving job swap past the job at the tail of the queue [40], Nudge- $K$  lets it pass up to  $K$  others while no job is passed more than once [41], with later refinements [42], and finite-skip models serve only among the oldest  $n$  waiting jobs [43]. The second is work:  $\gamma$ -Boost measures the work of later arrivals that are served ahead of a tagged job [17,18]. These analyses assume stationary Poisson arrivals and target means or tail asymptotics, not a cap enforced on every sample path. Our wrapper has the architecture of Load Balancing Guardrails [44], which leaves an arbitrary dispatching policy free except where a

constraint must be enforced, although that work optimises mean response time in heavy traffic. Additive bounds against a reference discipline are standard in packet scheduling [45–48], and the coefficient of Graham’s list-scheduling bound [49] has the same shape as the  $(2 - 2/k)$  of Section 6. Supplementary Section S9 compares these guarantees in detail.

### 2.3. Prediction-Driven Scheduling in Deployed Systems

Backfilling schedulers for high-performance computing replace users’ runtime estimates by predictions learned from past jobs, and report mean slowdown or mean wait [50–52]. Closest to our first case, Zhang, Wu and Lu predict the running time of submitted programs in a judging queue and order the jobs by prediction inside fixed-size windows [53], which limits overtaking by position; they report no per-job bound. Servers for large language models, which cannot cheaply interrupt a request, learn a relative order of the requests [54] or target the latency tail, because prediction-driven policies are fragile when the workload shifts [55]; others promote a request once its wait passes a threshold [56] or a multiple of its service time [57]. Serverless platforms approximate shortest remaining time first from user space [58] or learn a priority for each function [59]. None of these states a bound for individual jobs. A maximum waiting time does carry one, of the same form as ours (Section 6.6), and our guard can be placed around any of these schedulers.

### 2.4. Deadline-Driven Load

People concentrate their actions before a cutoff [2] and register in bursts before a deadline [3,4]. When the arrival rate changes as fast as jobs are served, congestion peaks after the arrivals do [60], and staffing rules need time-varying approximations [61]. These results size the capacity; they give no bound for an individual job once a queue has formed, and we keep the capacity fixed.

### 2.5. Relational Models for Cost Prediction

A job’s cost may depend on its user, its task and the relations between them, which is the setting of relational deep learning. On the RelBench benchmarks, graph models over linked tables beat gradient-boosted trees fitted on the columns of a single table [62], while on single tables tree ensembles still outperform deep models [63]. We therefore compare a heterogeneous graph neural network over users, tasks and assignments [20,64] with gradient-boosted trees [19] that carry engineered history features, and we judge them by the waits they produce in the queue as well as by their prediction scores.

## 3. Problem Model

### 3.1. Jobs, Servers and Dispatch

A finite set of  $n$  jobs is served by  $k$  identical servers. We say *server* for what a deployment calls an executor, a judging machine or a worker, and *job* for a submission, a function call or a test run. Job  $i$  arrives at time  $a_i \geq 0$  and needs  $C_i$  units of service, which we also call its *cost* or its *work*. Service is non-preemptive. A server that starts a job keeps it until the job finishes or has run for the time limit  $L$ , at which point the system stops it. Since  $C_i$  is the work actually executed,  $0 < C_i \leq L$ , and a job stopped at the limit contributes exactly  $L$ . For a stopped job, the waiting-time guarantee says nothing about the lost result. A *task* is the programming exercise a job attempts, so many jobs share one task.

Servers are work-conserving, which means that no server is idle while a job waits. Jobs are numbered in arrival order, with ties broken by input order, and we write  $j < i$  when  $j$  comes first in this order, so  $j < i$  implies  $a_j \leq a_i$ . At a common instant, completions are processed first, then arrivals, then dispatches. A *policy* is any rule that, whenever a server is free and a job waits, picks one waiting job and starts it. Such a rule may be

randomised, learned, priority-driven or driven by predictions of any quality. We make no stochastic assumption about the arrival times, the costs or their dependence, because the load we care about follows a calendar and is not stationary.

### 3.2. What the Scheduler Knows

A job's cost is hidden when it arrives, and the scheduler learns  $C_i$  only when  $i$  completes. At arrival it sees a feature vector  $x_i$ , built from the history of earlier jobs, and a score  $\hat{C}_i$  computed from  $x_i$ . This history must be causal. Write  $\text{done}_j$  for the instant at which the result of job  $j$  becomes readable, and  $d \geq 0$  for the delay before it reaches the store the features are read from. Then

$$\text{information about } j \text{ may enter } x_i \iff \text{done}_j + d \leq a_i. \quad (1)$$

In a simulation driven by a log,  $\text{done}_j$  can be read in two ways. One of the two ways takes the time at which the result appeared on the original platform, as the log records it. However, in the simulated queue the job completes at a time that depends on the policy being simulated. A feature that is causal on the original platform's clock may therefore use a result that the simulated queue has not produced yet. On the original clock we run the prediction study of Section 4, and on the simulated one the headline scheduling results (Section 8.1). The guarantees of Section 6 hold under either reading, since they assume nothing about the scores.

### 3.3. Waiting Time, the FCFS Reference and the Service Guarantee

Let  $s_i^P$  be the start time of job  $i$  under policy  $P$  and  $W_P[i] = s_i^P - a_i$  its waiting time. FCFS always starts the earliest-arrived waiting job, and when several servers become free at one instant it starts jobs in arrival order. We run FCFS on the same arrivals, costs and  $k$  as  $P$ , so that the two are compared job by job on one sample path. The *excess* of job  $i$  is

$$\text{excess}_P[i] = W_P[i] - W_{\text{FCFS}}[i]. \quad (2)$$

Writing  $j \prec i$  when  $j$  is started before  $i$ , the work that overtook  $i$  and the work that  $i$  overtook are

$$\text{In}_i = \sum_{j \succ i, j \prec i} C_j, \quad \text{Out}_i = \sum_{j \prec i, i \prec j} C_j. \quad (3)$$

A policy  $P$  offers a *service guarantee*  $G$  if  $\text{excess}_P[i] \leq G$  for every job  $i$ , on every arrival sequence, for all costs bounded by  $L$  and for every score, adversarial scores included. An operator states  $G$  in advance, and the guarantee is the proved statement that every job stays within it. This guarantee holds for each job on the realised sample path, not on average, and it assumes nothing about the quality of the score. Packet schedulers state additive per-flow bounds of this kind against a reference discipline [46]. Batch schedulers measure the same difference after the fact, against a fair start time [33]. However, the scheduling-with-predictions literature compares with an optimal policy [10,14]. We take FCFS as the reference because these services run it by default, so the excess is a quantity an operator can be held accountable for.

### 3.4. Metrics

The primary metric is the 99th percentile (p99) of the waiting time of the jobs that arrive in the 24 hours before a deadline, written  $\text{p99}_{\text{dl}}$ . This metric was fixed before any sealed part of a log was opened. With it we report the mean wait over all jobs and the largest excess, which is what a guarantee bounds. We also report the *harm*, the largest excess among the jobs that FCFS would have started within 1 s, that is, among jobs FCFS

would not have delayed. A fourth quantity, the *firing rate*, is the share of dispatch decisions at which the guard of Section 5 overrides the base policy, weighted by queue length. Finally, we report the worst wait among heavy jobs, defined in Section 4.

Improvements are expressed as the fraction of the gap closed,

$$\text{gap closed}(P) = \frac{M_{\text{FCFS}} - M_P}{M_{\text{FCFS}} - M_{\text{SJF}}}, \quad (4)$$

where  $M_Q$  is  $p99_{\text{dl}}$  under policy  $Q$  and SJF orders the waiting jobs by their true costs. SJF cannot be deployed, since costs are unknown at arrival. On several non-preemptive servers SJF also minimises none of our metrics, so it is a reference and not an optimum. A gap closed of 1 means that  $P$  matches the  $p99$  of SJF, and 0 that it matches FCFS. Gap closed is not a percentage reduction in waiting time, and the tables print the absolute waits beside it.

### 3.5. Steady-State Models and This Load

Stationary models give closed forms for the mean wait of each class of a non-preemptive priority queue [65, pp. 499–507]. The Schedule Ordered by Age-based Priority (SOAP) framework analyses every policy that ranks jobs by a function of their size class and age [66], a family that contains ranking by predicted cost. Both analyses need stationary arrivals, which deadline-driven load does not supply. On a synthetic Poisson stream the priority formula agrees with simulation to 0.4%, but on the same jobs at their recorded arrival times the measured mean wait is 10.7 to 13.7 times what the formula predicts (Supplementary Section S2.5). Thus, we use steady-state results as motivation only, and state every guarantee on the realised sample path.

## 4. Cost Prediction

The scheduler needs, for every arriving job, a number that says how long the job will run, computed from history alone. Users and tasks form a graph, because many users attempt the same task and each user attempts many tasks. A natural question is therefore whether a model that reads this graph predicts better than one that reads each job's own history. This section compares three kinds of model on the two grading logs of Section 7 and chooses the one the scheduler uses.

### 4.1. Targets and Measures

For ranking, the target is the conditional mean  $\mathbb{E}[C_i \mid x_i]$ . This mean orders a batch optimally on one server, by the pairwise-interchange argument of Section 5.1. The mix of jobs shifts between periods, so to compare predictors across periods we also score each model on a binary label. A job is *heavy* when its cost exceeds the 95th percentile of the training rows. On the main grading log this cut-off is 1.559 s, and heavy jobs are 1.31% of the test period against 5.00% of the training period. On the second log the cut-off is 1,314 ms and heavy jobs are 5.07%. For the heavy label we report the area under the receiver operating characteristic curve (AUROC). For the predicted cost we report the root mean squared error (RMSE) of  $\log(1 + C)$  and the Spearman rank correlation, since a ranking policy uses the order of the scores and not their level.

### 4.2. Protocol

Every model sees the same rows, the same split into training and test periods, and the same visibility rule (1). Historical features are computed by streaming the log in the order in which results became available and reading each user's and each task's state at the query instant. Recomputing these features from scratch gives 0 disagreeing rows in

all eight configurations we tested. Each test period is predicted by a model fitted only on earlier periods and frozen before the period's first arrival, and the heavy cut-off is refitted on that model's own training rows. Delaying the results by up to ten minutes before they may enter the features leaves the ranking intact and worsens only the calibration. Supplementary Section S1 gives these audits.

The three kinds of model are listed in Table 1, which compares what each representation knows. Every model in that table is a classifier for the heavy label, trained on the terms 2018-1 to 2019-2, tested on 2022-2 and averaged over 3 seeds. The scheduler uses a different fit on M4's features, the expected-cost regressor of Section 5.1, refitted on rolling forward periods (Supplementary Section S3.9).

- Gradient-boosted trees [19] over five feature groups, namely the task's history (M1), the user's history (M2), static features of the submitted code (M3), all of these together (M4), and M4 with one-hop aggregates over related users and tasks (M5).
- A GRU over the user's earlier jobs, used alone (R1) and as a state vector added to M4's features inside the trees (R1S).
- A heterogeneous GNN of the R-GCN type [20], over users, tasks, the assignments that group tasks and the class cohorts that group users. It is tested as a ladder of five models. The first is a multilayer perceptron on M4's features (N0). The others are the graph architecture with every neighbour set empty and no group nodes (G0U), with the group nodes but no edges (G0), with the real edges (G1), and with the edges shuffled within each period (G2). G2 keeps each node's number of neighbours and changes only who they are, so if the edges carry information, G2 must lose it.

**Table 1.** Predictors on the main grading log (CodeBench), 40,844 test rows. AUROC is for the heavy label, and Spearman is on the predicted cost. The second grading log and the remaining measures are in Supplementary Table S2.

| Kind  | Model | Information                          | AUROC  | Spearman |
|-------|-------|--------------------------------------|--------|----------|
| Trees | M1    | task history                         | 0.7960 | 0.219    |
|       | M2    | user history                         | 0.8591 | 0.302    |
|       | M3    | static features                      | 0.8632 | 0.290    |
|       | M4    | all of the above                     | 0.9283 | 0.312    |
|       | M5    | M4 and one-hop aggregates            | 0.9302 | 0.344    |
| GRU   | R1    | user's earlier jobs                  | 0.9239 | 0.313    |
|       | R1S   | M4 and the GRU state                 | 0.9366 | 0.329    |
| GNN   | N0    | M4, no graph                         | 0.9209 | 0.278    |
|       | G0U   | graph model, no edges or group nodes | 0.9348 | 0.321    |
|       | G0    | graph model, no edges                | 0.9325 | 0.324    |
|       | G1    | graph model, real edges              | 0.9324 | 0.329    |
|       | G2    | graph model, shuffled edges          | 0.9305 | 0.323    |

#### 4.3. What the Comparison Shows

The graph does not help on average. One-hop aggregates barely change the trees' AUROC, since M5 – M4 is +0.0019 of AUROC on the main log and –0.0002 on the second. In the GNN ladder, removing every edge (G0) or shuffling them (G2) leaves AUROC within 0.002 of G1. The paired intervals for G1 – G0, –0.0001 [–0.0019, 0.0018], and for G1 – G2, +0.0019 [–0.0000, 0.0042], both contain zero. The one step in the ladder that matters is N0 to G0U, +0.0139 [+0.0090, +0.0195] of AUROC, and this step changes the encoder while the edges stay empty.

The edges help only for tasks with little history of their own. On the 1,092 test rows whose task has fewer than five visible earlier results, G1 beats M4 by +0.1464 [+0.0997, +0.1887] in Spearman correlation. On these rows G1 also beats its own shuffled-

edge control by  $+0.0282$  [ $+0.0130, +0.0432$ ], whereas over all rows  $G1 - G2$  is  $+0.0063$  [ $+0.0014, +0.0111$ ]. We read this as a property of the task and not of relational models. When each user and each task already has a rich record, a neighbourhood adds little except for an entity with no record yet.

Prediction scores and queueing outcomes do not rank the models the same way. R1S has the highest AUROC, 0.9366. Its difference from M4's 0.9283 is  $+0.0083$  [ $+0.0038, +0.0131$ ]. Run through the queue with the same ranking objective as M4, however, R1S gains  $+0.007$  [ $0.003, 0.011$ ] of the gap at utilisation 0.8, loses  $-0.010$  [ $-0.022, -0.002$ ] at 0.5, and is unresolved at 1.0 (Section 7.2 defines the three loads). M5 has the higher AUROC of the two full tree models, and M5 loses in the queue at all three loads, by 0.012, 0.008 and 0.010 of the gap, each interval excluding zero. The scheduler therefore uses M4, chosen on the queueing outcome of validation periods rather than on prediction scores.

Which history matters varies by platform. The user's history leads on the main log, and the task's history leads on the second by 0.11 of AUROC (Supplementary Table S2). Supplementary Section S1.2 compares the feature groups on the four logs from outside education.

Prediction is cheap next to the queue. The median job on the main log runs for 204 ms, and charging each model's measured cost per job to the predictive policies changes the gap closed by less than 0.001 at every load (Supplementary Section S1).

## 5. Scheduling Algorithm

The scheduler has two layers. A base policy orders the queue by a score computed at arrival, and a guard limits how much work may overtake any waiting job. The guard accepts any base policy, and its guarantee does not depend on that policy behaving well.

### 5.1. Ranking by Expected Cost

When a server frees, the base policy starts the waiting job with the smallest score. For a batch on one server, the pairwise-interchange argument orders jobs by their conditional mean costs. However, this conclusion does not carry over to  $k \geq 2$  servers. On two servers, let job  $u$  cost 2 and job  $w$  cost 3, and let job  $v$  cost 0.1 with probability 0.9 and 100 with probability 0.1. Ordering by conditional mean starts  $u$  and  $w$  and gives an expected mean wait of  $2/3$ , whereas starting  $u$  and  $v$  gives 0.096667. We therefore treat expected-cost ordering as an empirically motivated base policy, not as a claim of optimality.

Squared error on raw seconds and the Tweedie deviance both have the conditional mean as their population optimum. Their fitted orders can nevertheless differ with finite model capacity, link function and regularisation, so we compare the objectives empirically, with the features and model of Section 4 held fixed. We fit the expected-cost policy, shortest predicted job first on the expected-cost score (SPJF-E), on raw seconds with a Tweedie loss of power 1.5, which accommodates mass near zero and a long right tail [67,68]. For the log-scale policy, shortest predicted job first on the log-scale score (SPJF-log), we fit the same pipeline by squared error on  $\log(1 + C)$ . By Jensen's inequality, the back-transform of this fit is no larger than the conditional mean.

We use only the order of a score. Its level does not transfer between periods, because the share of heavy jobs differs between training and test periods by a factor of several, so the scheduler never reads a score as a time or a threshold. Section 8.3 compares these and other objectives in the queue. No choice of score bounds the worst case. Whatever the score, a job it underestimates can be passed again and again, and only the guard below bounds that.

## 5.2. The Overtake-Budget Guard

Fix a base policy  $A$ . For every waiting job  $q$  and every time  $t$  let

$$\text{over}[q](t) = \sum \{ C_j : j > q, j \text{ has completed by } t \}, \quad (5)$$

the total cost of the jobs that arrived after  $q$  and have finished by  $t$ . Such a job ran while  $q$  waited, so it overtook  $q$ . Each waiting job has a budget  $\text{budget}(q, t)$  on this quantity, and the job is *eligible* when its account has reached its budget,  $\text{over}[q](t) \geq \text{budget}(q, t)$ . Whenever a server is free and the queue is not empty, the guard starts the earliest-arrived eligible job if there is one, and otherwise lets  $A$  choose (Algorithm 1 and Figure 2). Serving the earliest-arrived eligible job matters when budgets differ between jobs. It ensures that whenever a job  $j$  starts ahead of an earlier-arrived waiting job  $i$ , the job  $i$  was not eligible at that instant, and the per-job bound of Section 6 rests on this step. With a constant budget  $B$  the account is non-increasing in arrival order, so the rule reduces to a test on the head of the queue, and  $B = 0$  gives FCFS.

The guarantee reads the budget only through an upper bound on it. If  $0 \leq \text{budget}(q, t) \leq B_{\max}$  for every waiting job at every instant, then every job waits at most  $W_{\text{FCFS}} + B_{\max}/k + (3 - 2/k)L$ , whatever the base policy, the arrivals and the predictions (Theorem 7). Below the cap, the budget decides how much of the base policy's benefit survives and which jobs pay for it. We study the family

$$\text{budget}(q, t) = \min(B_0 + \gamma n_q + \eta k(t - a_q), B_{\max}), \quad (6)$$

with  $B_0, \gamma \geq 0$ ,  $0 \leq \eta < 1$  and  $n_q$  the number of jobs waiting when  $q$  arrives. We study three shapes in this family, namely *constant* ( $\gamma = \eta = 0$ ), *age-relative* ( $\eta > 0$ ) and *queue-length* ( $\gamma > 0$ ). In the age-relative shape,  $\eta$  is the share of the system's capacity, over a job's own wait, that may go to jobs that arrived after it, so the allowance grows with the wait the job has already had. The term  $\gamma n_q$  widens the allowance only for jobs that arrive into a long queue. All of  $B_0$ ,  $\gamma$  and  $B_{\max}$  are measured in work and scale with  $k$ .

---

### Algorithm 1 Overtake-budget guard wrapped around a base policy $A$

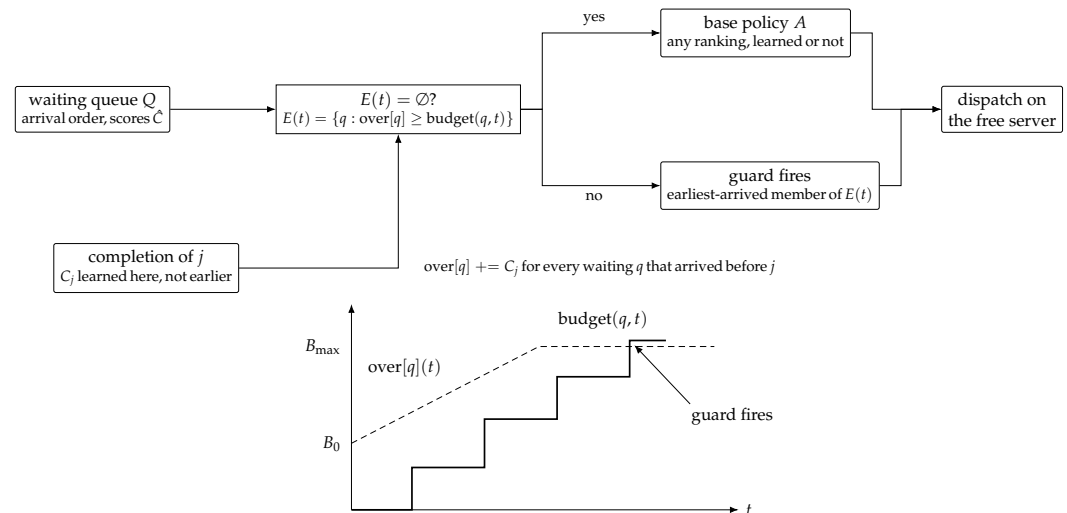
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```

1: parameters: cap  $B_{\max} \geq 0$ ; shape  $B_0, \gamma \geq 0, \eta \in [0, 1]$ ; base policy  $A$ 
2: state: waiting set  $Q \leftarrow \emptyset$ ;  $\text{over}[q] \leftarrow 0$  for every job  $q$ 
3: while some job has not started do
4:    $t \leftarrow$  next event time
5:   for all jobs  $c$  completing at  $t$  do  $\triangleright C_c$  becomes known here and nowhere earlier
6:     for all  $q \in Q$  that arrived before  $c$  do
7:        $\text{over}[q] \leftarrow \text{over}[q] + C_c$ 
8:     end for
9:   end for
10:  for all jobs  $i$  arriving at  $t$  do
11:     $n_i \leftarrow |Q|$ ;  $Q \leftarrow Q \cup \{i\}$ ;  $\text{over}[i] \leftarrow 0$ 
12:  end for
13:  while a server is free and  $Q \neq \emptyset$  do  $\triangleright$  dispatch decision
14:     $E \leftarrow \{ q \in Q : \text{over}[q] \geq \min(B_0 + \gamma n_q + \eta k(t - a_q), B_{\max}) \}$ 
15:    if  $E \neq \emptyset$  then
16:       $j \leftarrow$  the earliest-arrived member of  $E$   $\triangleright$  guard fires
17:    else
18:       $j \leftarrow A(Q, t)$   $\triangleright$  base policy decides
19:    end if
20:    start  $j$  on the free server;  $W[j] \leftarrow t - a_j$ ;  $Q \leftarrow Q \setminus \{j\}$ 
21:  end while
22: end while

```

---



**Figure 2.** The guard of Algorithm 1 wrapped around a base policy (above), and the account it keeps for one waiting job  $q$  (below). Only completed work is charged, because a job's cost is unknown until it finishes, so  $\text{over}[q]$  rises by one step at each completion of a job that arrived after  $q$ . When the account reaches the budget,  $q$  becomes eligible, and the guard starts the earliest-arrived eligible job, which need not be  $q$ . The budget drawn is the age-relative shape, rising from  $B_0$  at rate  $\eta k$  and capped at  $B_{\max}$ ; the bound depends on the cap alone, so any budget below the dashed ceiling carries the same guarantee. The lower panel is a schematic.

### 5.3. Setting the Guard

An operator states a bound  $G$ , the number of seconds by which any job may exceed its FCFS wait, and sets  $B_{\max} = k(G - (3 - 2/k)L)$  (Corollary 8). No data are needed for this step. A bound of at most  $(3 - 2/k)L$  can be guaranteed this way only with  $B_{\max} = 0$ , which is FCFS. The shape parameters  $B_0$ ,  $\gamma$  and  $\eta$  do not affect the bound, and we choose them on validation data by a rule fixed in advance (Section 7.3). At  $G = 600$  s this rule selects the age-relative shape with  $B_0 = 30k$  s of work and  $\eta = 0.75$  (Supplementary Table S5).

The guard charges the executed cost of an overtaker when it completes, because a job's cost is known only then. A job still running contributes nothing yet. When the last overtaker of a job  $i$  starts, up to  $k - 1$  earlier overtakers may still be running, and the one starting is not charged either. The work that actually passes  $i$  can therefore exceed the budget by at most  $kL$ . This is what the additive term of the guarantee pays for, and it is why the guard needs no prediction to enforce its bound. With a Fenwick tree over completed work and a segment tree over the waiting jobs, each event costs  $O(\log n)$ , and all budget arithmetic is done in integer microseconds, so the eligibility test is exact (Supplementary Section S5).

### 5.4. Compared Policies

Table 2 names the policies compared in Section 8. Our method is Guard( $G$ ), the guard with bound  $G$  wrapped around SPJF-E. The other rows separate its parts or test it.

Every simulation that includes a guard checks the per-job bound of Section 6 on every job while it runs (Section 8.5).

## 6. Theory

Everything in this section is stated for one fixed run: one arrival sequence, one vector of costs and one realisation of whatever randomness the base policy uses. No stationarity, no independence and no assumption about the scores enters a proof, and the upper

**Table 2.** Policies compared in the experiments.

| Name                                 | Role                                                                                                    |
|--------------------------------------|---------------------------------------------------------------------------------------------------------|
| FCFS                                 | Deployed baseline and the reference of the guarantee                                                    |
| SJF                                  | Orders by true cost, cannot be deployed, and is the reference of the gap closed                         |
| SPJF-E                               | Orders by the expected-cost score of Section 5.1, unguarded                                             |
| SPJF-log                             | Orders by the log-scale score, unguarded                                                                |
| Guard(G)                             | SPJF-E inside the guard with bound $G$ and the selected shape (the method)                              |
| Guard-fixed(G),<br>age(G), -queue(G) | - The same guard with the best constant, age-relative or queue-length shape                             |
| Fixed(G)                             | The same guard with the whole cap as budget, budget $\equiv B_{\max}$                                   |
| Skip(G)                              | A guard that counts overtakes instead of work, charged at start, at the count that gives bound $G$      |
| Timeout( $\theta$ )                  | A maximum waiting time, which serves the earliest-arrived job once it has waited $\theta$ (Section 6.6) |
| Corrupted scores                     | SPJF-E and Guard(G) under a reversed, a random and an inverted-tail score                               |

bounds hold even against a base policy that knows the future. An identity ties a job's delay relative to FCFS to the work that overtook it (Sections 6.2–6.4), two bounds turn it into the guarantee of Algorithm 1 (Section 6.5), and three further results cover a maximum waiting time, an absolute bound and completion reports that arrive late (Sections 6.6–6.8). Full proofs and the extremal constructions are in Supplementary Sections S4 and S8.

### 6.1. Conventions

We use the model of Section 3 and fix three readings the proofs depend on. The jobs started at one instant form a *phase*, ordered as the policy starts them, and  $j \prec i$  refers to the concatenation of the phases. Within a phase, FCFS starts the jobs in arrival order. A later-arriving job started in the same phase as  $i$  but ahead of it counts as overtaking  $i$  and contributes its whole  $C_j$  to  $\text{In}_i$ , although it executes nothing while  $i$  waits. Remaining work at  $a_i$  is read after the arrivals at  $a_i$ .

### 6.2. The Workload Gap between Two Policies

Write  $U_Q(t)$  for the unfinished work at time  $t$  under policy  $Q$ , counting every job with  $a_j \leq t$ .

**Lemma 1** (Workload comparison). *Let  $Q_1$  and  $Q_2$  be non-preemptive work-conserving  $k$ -server policies run on the same input. Then  $|U_{Q_1}(t) - U_{Q_2}(t)| \leq (k-1)L$  for every  $t$ . For  $k \geq 2$  the inequality is strict,  $|U_{Q_1}(t) - U_{Q_2}(t)| < (k-1)L$ , and the constant is the exact supremum, approached but never attained. At  $k = 1$  both sides vanish, because one server executes work at rate 1 whenever at least one job is present, so every work-conserving policy has the same unfinished work at every instant.*

**Proof sketch of Lemma 1.** The difference  $D = U_{Q_1} - U_{Q_2}$  is continuous and piecewise linear. Whenever  $U_{Q_1}$  exceeds  $(k-1)L$ , more than  $k-1$  jobs are present under  $Q_1$ , every server of  $Q_1$  is busy, and  $D$  cannot increase. Tracking  $D$  back to the last instant at which  $U_{Q_1}$  was at most  $(k-1)L$  gives the bound. Supplementary Sections S4.1 and S8.1 give the proof, the strictness and the construction that shows the constant cannot be lowered.  $\square$

The argument that a work-conserving  $k$ -server system can fall behind only by what  $k-1$  servers hold is the one used to compare a multiserver system with a single fast server in the analysis of shortest remaining processing time (SRPT) [69,70], where it bounds the

work below a size cutoff. Here we use it to compare any two non-preemptive work-conserving policies over all of the work. FCFS also minimises the unfinished-work vector in a stronger order, but only among policies whose assignment of jobs to servers ignores their costs [71], which the policies compared here do not.

### 6.3. The Net-Overtake Identity

For a job  $i$  and a policy  $Q$ , let  $R_i^Q$  be the remaining work at time  $a_i$ , under  $Q$ , of the jobs that arrived before  $i$ , and let  $\beta_i^Q$  be the remaining work at  $s_i^Q$  of the jobs started before  $i$  that are still running then. Write

$$\Delta_i = k(W_P[i] - W_{\text{FCFS}}[i]) - (\text{In}_i - \text{Out}_i). \quad (7)$$

**Theorem 2** (Net-overtake identity). *For every non-preemptive work-conserving  $k$ -server policy  $P$ , every input and every job  $i$ ,*

$$\Delta_i = (R_i^P - R_i^{\text{FCFS}}) - \beta_i^P + \beta_i^{\text{FCFS}}, \quad |\Delta_i| \leq 2(k-1)L, \quad (8)$$

with strict inequality for  $k \geq 2$ . Equivalently,  $|W_P[i] - W_{\text{FCFS}}[i] - (\text{In}_i - \text{Out}_i)/k| \leq (2 - 2/k)L$ . At  $k = 1$  the error term vanishes and  $W_P[i] - W_{\text{FCFS}}[i] = \text{In}_i - \text{Out}_i$  exactly.

**Proof sketch of Theorem 2.** Job  $i$  waits throughout  $[a_i, s_i^P]$ , so every server is busy there and exactly  $kW_P[i]$  units of work are executed, none of it on  $i$ . Splitting that work by arrival order gives

$$kW_P[i] = R_i^P + \text{In}_i - \text{Out}_i - \beta_i^P. \quad (9)$$

The jobs counted in  $\beta_i^P$  run on distinct servers other than the one that takes  $i$ , so  $0 \leq \beta_i^P \leq (k-1)L$ . Under FCFS,  $\text{In}_i = \text{Out}_i = 0$ , and Lemma 1 applied at  $a_i$  bounds  $|R_i^P - R_i^{\text{FCFS}}|$  by  $(k-1)L$ . Subtracting the two instances of (9) gives the identity. Each direction uses only two of the three ranges, which is why the slack is  $2(k-1)L$  and not  $3(k-1)L$ . Supplementary Section S4.3 gives the proof in full.  $\square$

**Corollary 3** (A work budget is sufficient). *If a policy keeps  $\text{In}_i - \text{Out}_i \leq Z$  for every job, in particular if it keeps  $\text{In}_i \leq Z$ , then  $\text{excess}_P[i] \leq Z/k + (2 - 2/k)L$  for every job.*

**Proof of Corollary 3.** By Theorem 2,  $k \text{excess}_P[i] \leq \text{In}_i - \text{Out}_i + 2(k-1)L \leq Z + 2(k-1)L$ . Dividing by  $k$  gives the claim.  $\square$

So a scheduler that never lets more than  $Z$  units of work pass a waiting job holds every wait within  $Z/k + (2 - 2/k)L$  of its FCFS wait, whatever it ranks by. The constant cannot be improved.

**Theorem 4** (Tightness of Theorem 2). *For every fixed  $k \geq 2$ , service bound  $L > 0$  and  $\varepsilon > 0$  there are inputs, work-conserving policies  $P$  and jobs  $i$  with  $\Delta_i > 2(k-1)L - \varepsilon$ , and there are others with  $\Delta_i < -2(k-1)L + \varepsilon$ . Hence  $\sup |\Delta_i| = 2(k-1)L$ , a supremum that is never attained.*

**Proof sketch of Theorem 4.** Both witnesses start from a geometric cascade that drives  $U_P - U_{\text{FCFS}}$  to  $(k-1)L(1 - ((k-1)/k)^m)$  in  $m$  rounds and then place a victim so that FCFS starts it at once while  $P$  holds it back (upward), or the mirror image (downward). The instances need hundreds of jobs and are rescaled to keep  $L$  fixed. Supplementary Section S8.2 gives both in full. Non-attainment follows from Lemma 1.  $\square$

In the downward direction, half of the extremal value comes from the convention of Section 6.1 that charges a same-phase overtaker in full. Supplementary Section S4.3 gives the variant that charges executed work only.

#### 6.4. A Work Budget Is Also Necessary

A policy that guarantees anything about individual jobs is forced to bound the work that overtakes them.

**Lemma 5** (Outbound work under a guarantee). *Let  $i$  be a job, and suppose  $\text{excess}_P[i] \leq G$  and  $\text{excess}_P[j] \leq G$  for every job  $j$  counted in  $\text{Out}_i$ . Then  $\text{Out}_i \leq k(G - \text{excess}_P[i] + L)$ .*

**Proof sketch of Lemma 5.** Every job counted in  $\text{Out}_i$  starts inside a window of length  $G - \text{excess}_P[i]$ , and of the jobs one server starts inside such a window all but the last complete inside it (Supplementary Section S4.7).  $\square$

**Corollary 6** (A work budget is necessary). *If  $P$  satisfies  $\text{excess}_P[j] \leq G$  for every job on some input, then  $\text{In}_i \leq kG + (3k - 2)L$  for every job  $i$ .*

**Proof of Corollary 6.** Theorem 2 gives  $\text{In}_i \leq k \text{excess}_P[i] + \text{Out}_i + 2(k - 1)L$ . Lemma 5 bounds  $\text{Out}_i$ , and the terms in  $\text{excess}_P[i]$  cancel.  $\square$

The two corollaries say that bounding  $\text{In}_i$  and offering a per-job guarantee are the same requirement, up to an additive  $O(kL)$ . A bound on the number of overtakes, as in the mechanisms of Section 2, is not necessary for a guarantee, and it gives one only by charging each pass the worst case  $L$ . That charge makes it looser than a work budget by the ratio of  $L$  to the mean cost, 150 on the main grading log ( $L = 60$  s against a mean cost of 0.40 s). Supplementary Section S4.4 proves both halves and gives the conversion from a count to a bound.

#### 6.5. The Guarantee of the Guard

Nothing below restricts the base policy, which may be randomised, learned or clairvoyant, and nothing restricts the budget rule beyond the cap: it may depend on the job, decrease over time, and take different values at two decisions made at the same instant.

**Theorem 7** (Guard bound). *Fix a base policy  $A$ , a cap  $B_{\max} \geq 0$ , and a budget rule that respects the cap, that is,  $0 \leq \text{budget}(q, t) \leq B_{\max}$  for every waiting job  $q$  and every dispatch decision at time  $t$ , and run Algorithm 1 with that rule. For every input and every job  $i$ , writing  $W = W_P[i]$  for the wait under the guard,*

$$\text{In}_i < B_{\max} + kL, \quad (10)$$

$$W < W_{\text{FCFS}}[i] + B_{\max}/k + (3 - 2/k)L, \quad (11)$$

where (10) is asserted whenever  $i$  has at least one overtaker.

**Proof sketch of Theorem 7.** When the last overtaker of  $i$  starts,  $i$  is waiting and not eligible, so  $\text{over}[i]$  is below its budget there. The overtakers started earlier and not yet finished occupy at most  $k - 1$  servers, and the last one adds a further  $L$ , which gives (10). Theorem 2 with  $\text{Out}_i \geq 0$  turns it into (11). Supplementary Section S4.5 gives the proof in full.  $\square$

**Corollary 8** (The cap for a given bound). *Setting  $B_{\max} = k(G - (3 - 2/k)L)$  makes Algorithm 1 offer the service guarantee  $G$ , whatever the budget shape below the cap and whatever the*

base policy  $A$ . The choice is available when  $G \geq (3 - 2/k)L$ , and gives a positive budget only when the inequality is strict. At equality  $B_{\max} = 0$  and the guard is FCFS.

**Proof of Corollary 8.** Substitute  $B_{\max}$  into (11).  $\square$

The proof reads the budget only through the cap, so the shape below the cap is free. The age-relative shape with one constant  $B_0$  for every job also gives a multiplicative bound,  $(1 - \eta)W < W_{\text{FCFS}}[i] + B_0/k + (3 - 2/k)L$ . The queue-length shape does not, by a margin that grows with  $\gamma$  (Supplementary Sections S4.5 and S4.6). Neither constant of Theorem 7 can be improved: with a constant budget the term  $B_{\max}/k$  is reached exactly, and  $(3 - 2/k)L$  is the exact supremum at every  $k$ , approached by a cascade (Supplementary Section S8.3). If the service enforces a run-time cap known at arrival, charging it at the start and refunding the difference at completion removes  $L$  from the constant (Supplementary Section S7.6). Finally, a guard that never fires leaves the base policy's run unchanged, which can be checked on that run before the guard is deployed (Supplementary Section S4.2).

#### 6.6. A Maximum Waiting Time Gives the Same Guarantee

Deployed schedulers guard against starvation with a maximum waiting time, and recent inference servers promote a request once its wait passes a threshold [56,57], with no stated bound. In the simplest form, once the earliest-arrived waiting job has waited  $\theta$  it is started next, and until then the base policy decides. We call this rule Timeout( $\theta$ ). It reads arrival times and a clock, never a cost.

**Proposition 9** (Timeout bound). Fix a base policy and  $\theta \geq 0$ . For every input and every job  $i$  with at least one overtaker, Timeout( $\theta$ ) gives  $\text{In}_i < k(\theta + L)$ , and every job waits

$$W < W_{\text{FCFS}}[i] + \theta + (3 - 2/k)L. \quad (12)$$

**Proof sketch of Proposition 9.** From  $a_i + \theta$  on, the earliest-arrived waiting job has waited at least  $\theta$ , so every job started before  $s_i$  arrived before  $i$ . Each overtaker of  $i$  therefore starts before  $a_i + \theta$  and runs at most  $L$ , and on each server the overtakers fit in  $[a_i, a_i + \theta + L)$ . Theorem 2 with  $\text{Out}_i \geq 0$  gives the bound (Supplementary Section S4.11).  $\square$

At  $\theta = G - (3 - 2/k)L$  the clock gives the bound that the guard gives at  $B_{\max} = k\theta$ . Unlike Theorem 7, the constant of Proposition 9 is not shown to be tight: the worst ratio of excess to bound found by search is 0.996 at  $k = 1$  and 0.756 to 0.829 for  $k = 2$  to 4 (Supplementary Section S4.11), so Section 8.4 also compares the two rules at equal realised worst excess. The rules differ in what they charge a waiting job for. The clock charges it for all the work the servers do while it waits, the guard only for completed work of jobs that arrived after it. Near saturation, a job that arrives behind a backlog is charged for the backlog, the clock fires for most of the queue, and the schedule falls back to arrival order at the loads where ordering by cost pays most.

The two rules can be combined: start the earliest-arrived job if it has waited  $\theta$ , and otherwise run Algorithm 1. Both bounds then hold at once. The order matters, since serving the guard's eligible job first can pass over a job that has already waited  $\theta$  (Supplementary Section S4.11). The clock's half reads no completion report, so the combined rule keeps (12) when reports arrive late or never.

#### 6.7. An Absolute Bound from the Arrival Log

A bound relative to FCFS is a margin over a schedule that is not run. The arrival log alone gives an absolute bound, which an operator can state as a number of seconds. Let

$V_i^-$  be the work that a single server of rate  $k$ , fed the same jobs, would still hold at  $a_i$  from the jobs that arrived before  $i$ :  $V_1^- = 0$  and  $V_i^- = \max(0, V_{i-1}^- + C_{i-1} - k(a_i - a_{i-1}))$ . The largest value  $\sigma = \max_i(V_i^- + C_i)$  is the depth of the smallest token bucket of rate  $k$  that the arrivals, counted in work, conform to, which is the burst parameter of network calculus [72,73].

**Proposition 10** (Absolute bound). *For every non-preemptive work-conserving policy and every job  $i$ ,  $k W[i] \leq V_i^- + (k-1)L + \text{In}_i$ . Hence  $W_{\text{FCFS}}[i] \leq (V_i^- + (k-1)L)/k$ , and under Algorithm 1 with cap  $B_{\max}$*

$$W < \frac{V_i^- + B_{\max}}{k} + \left(2 - \frac{1}{k}\right)L \leq \frac{\sigma + B_{\max}}{k} + \left(2 - \frac{1}{k}\right)L. \quad (13)$$

**Proof sketch of Proposition 10.** Equation (9) gives  $k W[i] \leq R_i + \text{In}_i$ . A work-conserving  $k$ -server queue never holds more than  $(k-1)L$  above the single fast server: whenever it does, all  $k$  servers are busy and it drains at the same rate. Hence  $R_i \leq V_i^- + (k-1)L$ , and Theorem 7 bounds  $\text{In}_i$  (Supplementary Section S4.11).  $\square$

At  $k = 1$  the fast server is the queue itself,  $V_i^- = W_{\text{FCFS}}[i]$ , and the proposition is Theorem 7 again. An operator who guarantees that no job waits longer than  $D$ , over a period whose arrivals fit a bucket of depth  $\sigma$ , sets  $B_{\max} = kD - \sigma - (2k-1)L$ : the capacity over the deadline, less the burst, less  $2k-1$  time limits. The cap is non-negative only when  $D \geq \sigma/k + (2-1/k)L$ , and a shorter deadline cannot be guaranteed this way. The clock obeys the same bound with  $k\theta$  in place of  $B_{\max}$ . Supplementary Section S4.11 relates the statement to earlier sample-path comparisons with a fast server, some of which were shown to be false [74].

#### 6.8. Completion Reports That Arrive Late

Theorem 7 assumes that a completion is charged before the next dispatch decision. In a distributed service the servers are separate machines that report their completions to the scheduler over a network, and a server may take its next job before its report has been counted. Suppose every report reaches the scheduler's counter within  $\delta$  of the completion, while servers still take a job the instant they free.

**Proposition 11** (Late charges). *Under Algorithm 1 with a shared counter and charges delayed by at most  $\delta$ , every job  $i$  with an overtaker has  $\text{In}_i < B_{\max} + kL + k(L + \delta)$ , and*

$$W < W_{\text{FCFS}}[i] + B_{\max}/k + (4 - 2/k)L + \delta. \quad (14)$$

*For  $\delta > 0$  the bound is attained in the limit at  $k = 1$  and at  $k = 2$ . If each server applies its own completion report in the same step as its next pull, and only other servers' reports travel late, Theorem 7 holds unchanged for every  $\delta$ , lost reports included.*

**Proof sketch of Proposition 11.** When the last overtaker starts, the charged part is below the budget as before. The uncharged overtakers are those still running, as in Theorem 7, and those that finished within the last  $\delta$  and whose report has not arrived. On one server the latter are at most one job that started before that window and jobs that fit inside it, so they hold less than  $L + \delta$ . When a server reports before it takes its next job, an uncounted completion can belong only to a server that finished at the decision instant itself and has not yet taken a job, so it holds nothing in service. Supplementary Section S4.12 gives both proofs and the instances that attain the bound.  $\square$

At  $k = 1$  any  $\delta > 0$  already costs a whole  $L$ , because the single server takes its next job the instant it finishes one, before that job's report can arrive. The remedy is a protocol change rather than a smaller budget: a server that counts its own completion before taking the next job restores Theorem 7, and the combined rule of Section 6.6 keeps its clock bound for any delay.

#### 6.9. Which Assumptions Can Be Dropped

We may replace the global bound  $L$  by the largest cost seen so far, read at the right instant. Servers may run at different speeds, with the budget divided by total capacity, and a fixed setup paid before every job is absorbed into its cost. Work conservation cannot be dropped. If a server may idle while jobs wait, the identity fails by exactly the forced idle time. Supplementary Section S4.10 states these relaxations with their sharp forms.

#### 6.10. A Floor of Order $L$ , and What a Guarantee Costs

No budget removes the last  $L$  of the additive constant, because a job can be displaced before anything has completed, and at that instant a guard that charges completed work has nothing to charge.

**Proposition 12** (Floor). *Fix a budget rule that assigns a positive budget to every waiting job at the first dispatch decision. Then there are a base policy  $A$  that observes a job's cost only at its completion, and an input, such that Algorithm 1 run around  $A$  with that rule leaves some job with excess  $\geq L$ . Consequently no bound  $G < L$  can be guaranteed by a wrapper of this family that holds for every base policy, or for one prediction-ranked base policy under arbitrary predictions, which is the generality in which Theorem 7 is stated.*

**Proof sketch of Proposition 12.** Put  $k + 1$  jobs of cost  $L$  into an empty system at  $t = 0$ . Nothing has completed, so no job is eligible, the guard accepts the base policy's choices, and a base policy that starts the last-arrived job in the first phase pushes back a job that FCFS starts at once, by  $L$ . Supplementary Section S4.8 gives the proof and shows that the hypothesis can be neither dropped nor weakened.  $\square$

At one server the floor is the whole story: with a constant budget  $B > 0$  the bound  $B + L$  of Theorem 7 is the exact supremum of the excess (Supplementary Section S8.4). The guarantee also caps how much reordering can gain at all.

**Theorem 13** (Price of the guarantee). *Take  $m$  jobs of cost  $L$  and  $n$  jobs of cost  $c < L$ , the long ones arriving first, all at  $t = 0$ , and let  $P$  be a non-preemptive work-conserving policy with excess $_P[j] \leq G$  for every job. Define  $D_0 = mn(L - c)/k$ ,  $H = 2(1 - 1/k)L(m + n)$  and  $\alpha = (kG + (3k - 2)L)/(nc)$ . If  $D_0 > H$ , equivalently  $mn(L - c) > 2(k - 1)L(m + n)$ , then*

$$\frac{\overline{W}_{\text{FCFS}} - \overline{W}_P}{\overline{W}_{\text{FCFS}} - \overline{W}_{\text{SJF}}} \leq \alpha + \varepsilon_+, \quad \varepsilon_+ = \frac{H(1 + \alpha)}{D_0 - H}. \quad (15)$$

For fixed  $k, L, c$  and  $G$ , the finite-sample term is  $O((m + n)/(mn))$ .

**Proof sketch of Theorem 13.** Count the inverted pairs of long and short jobs in two ways, once through the mean waits, which reads the interchange argument behind Smith's rule [75] on this input, and once through Corollary 6 applied to each long job. Supplementary Section S8.5 gives the proof.  $\square$

The bound is capacity times  $G$ , divided by the work that wants to overtake. It is linear in  $G$  and reaches 1 when  $kG$  covers the total cost of the short jobs, and because

it applies to every policy with the guarantee, it describes the trade-off an operator faces rather than a weakness of this guard. At  $k = 1$  the achievable fraction is known exactly,  $\lfloor G/c \rfloor / n$  capped at 1, and the guard around exact SJF meets the bound of Theorem 13 on this family up to  $2(3k - 2)L/(nc)$  and finite-sample terms (Supplementary Sections S8.5 and S8.6, with an exhaustive check over 1,161 parameter settings).

## 7. Workloads and Experimental Setup

### 7.1. The Workloads

We use six logs of real jobs and one log of student activity (Table 3). Our main log comes from CodeBench, the programming-course platform of the Universidade Federal do Amazonas in Brazil, which records every run of a student's code together with its duration [21]. A second grading log comes from ACcoding, an online judge used in university programming courses [22]. ACcoding's public release has no timestamps, only submission identifiers. Without timestamps, we keep the real order and costs of its jobs and draw their arrival times, so its results test the ordering mechanism but not waiting in seconds.

Four logs come from outside education. The Azure Functions trace of 2021 covers two weeks of calls to functions that run on demand [5,6]. Two pools of test machines that Mozilla uses for the Firefox browser, which we call CI pools A and B, recorded the waits we use to check our simulator [23,24]. In Intel's Netbatch compute farm [25,26], single jobs run for days, and this farm is the case where our method does not apply.

Finally, the Open University Learning Analytics Dataset (OULAD) records the daily activity of students in online courses [76]. Because it contains no jobs, we use it only to check that activity crowds before deadlines outside our own grading logs.

**Table 3.** The logs, what each records and what we use it for. "Arrivals" says whether arrival times are recorded, reconstructed from the recorded end time and duration, or drawn; "Waits" whether each job's queueing wait is recorded.

| Log                      | Use                                    | Size                                                                 | Arrivals      | Waits |
|--------------------------|----------------------------------------|----------------------------------------------------------------------|---------------|-------|
| CodeBench [21]           | main scheduling log                    | 4,611,325 runs; 863,149 graded submissions in the development period | reconstructed | no    |
| ACcoding [22]            | ordering on a second grading platform  | 4,046,652 submissions, 3,237,322 outside the sealed block            | drawn         | no    |
| Azure Functions 2021 [5] | a cloud workload                       | 1,980,951 calls over 14 d                                            | reconstructed | no    |
| CI pool A [23,24]        | simulator check; light tail            | 191,970 runs over 21 d                                               | recorded      | yes   |
| CI pool B                | second simulator check                 | 30,702 runs over 7.0 d                                               | recorded      | yes   |
| Intel Netbatch [25,26]   | a case where the method does not apply | 9,054,066 jobs over 30 d, 8,441,207 serial with positive run time    | recorded      | yes   |
| OULAD [76]               | deadline crowding elsewhere            | 9 course presentations, daily counts                                 | —             | —     |

Three features of the main log shape how we use it, and Supplementary Section S2 gives the full audit. Since the timestamp of a run is the moment its result was written, we reconstruct the arrival as  $a_i = t_i - C_i$  and treat the recorded stamp as  $\text{done}_i$ . This is the time the run started. CodeBench runs each student's code in its own container (Section 7.2), so we take it as the submission time, and the log does not record the delay between the two. Supplementary Section S3.15 reads the stamp the other way, as the time of receipt, for an earlier configuration of the method.

Stamps are whole seconds while durations are in microseconds, which lets a model read the fractional part of the gap to the previous record and learn from it something it should not know. Given only that part, a classifier separates heavy from light jobs with

AUROC 0.8091. This AUROC falls to 0.4499, which is chance, when each stamp is jittered deterministically, and the forward models move by less than 0.008.

No run after 2019-08-28 lasts longer than 60 s, although the platform documents no timeout. Hence we take the periods after that date, cap each duration at the observed ceiling,  $C^{\text{cap}} = \min(C, 60)$ , and impose  $L = 60$  s in our pool, as a shared service would (Supplementary Table S3). The logs from outside education document no time limit. For them we take  $L$  to be the 99.9th percentile of running times in the log's training period and cap longer jobs at it, an analysis choice that a deployment would have to enforce by stopping those jobs (Supplementary Table S3).

## 7.2. A Shared Pool Built from the Log

CodeBench runs each student's code in a separate container [77], so its log records how long each run took but not a shared queue. Our experiments therefore replace the containers with a pool of  $k$  servers fed by the recorded runs, a queue of our own construction.

Three things support this construction. A fixed pool of judging machines on one queue is a standard deployment [7,8]. On the same demand, one container per student keeps 1,636 containers alive at the p99, and each of them runs code for 0.12% of its lifetime, whereas a pool of 14 servers holds the p99 of waits before deadlines under 1 s. Finally, the simulator reproduces waits that someone else recorded. When fitted on one part of each CI log, it returns the other part's mean wait, hourly profile and per-job order (Supplementary Section S2.3). None of this shows that the original platform ever queued, and we make no claim about waiting there.

A single course in a single term does not load even one server, so we overlay several. From the 40 class-terms of the development period, each holding the submissions of one class in one academic term, we superimpose 44 copies. The copies are aligned by weekday and hour, and each is shifted by a whole number of weeks drawn at random. Overlaying whole terms keeps the daily and weekly cycles and each job's position relative to its own deadline, which compressing the gaps between arrivals would not. In all, 17,634,760 jobs arrive over 30 weeks, and 4,219,556 of them arrive in the 24 hours before an assessment deadline.

Three loads share this trace and differ only in the number of servers. Four of the five overlays, which differ only in the drawn week shifts, use  $k = 8, 5$  and  $4$ . On the fifth, whose busy hours carry the least work,  $k$  is  $7, 5$  and  $4$ , so tables that average over overlays print  $k = 7.8$  at the lightest load. Averaged over the five overlays, the utilisation of the busiest hours is 0.502, 0.783 and 0.979, and we call the three loads  $\rho = 0.5, 0.8$  and  $1.0$ . These loads are 0.529, 0.847 and 1.059 on the first overlay alone.

The five overlays are one construction and not five platforms. For intervals we use 95% percentile intervals from 2,000 resamples of whole weeks, drawn once and shared by every policy so that differences between policies are paired. Such intervals cover variation over time, not the choice of terms or courses. Resampling whole class-terms instead and rebuilding the overlays each time widens them several-fold. At the busiest load Guard(600) then closes 0.801 of the gap with interval [0.328, 0.913], and its difference to unguarded ordering still excludes zero (Supplementary Table S18).

Arrivals follow the course calendar. When the number of runs per class is predicted a week ahead, the log-scale RMSE is 1.243 from recent counts alone, 1.159 from calendar features alone and 1.112 from both. Adding the calendar thus changes the error by  $-0.1307$  [ $-0.1395, -0.1213$ ].

OULAD shows the same pattern outside our grading logs. In each of its nine course presentations, the three days before an assessment carry 1.29 to 2.95 times the activity of

other days. With calendar features, a warning of peak activity seven days ahead reaches an F1 score (the harmonic mean of precision and recall) of 0.593, against 0.000 for a moving average of the previous two weeks.

### 7.3. Development Data, Sealed Data and Parameter Selection

Three logs have a part that was sealed before the method was designed: three academic terms of CodeBench (2023-1, 2023-2 and 2024-1), the last 20% of ACcoding's submission identifiers, and OULAD's 2014 year. Of the 809,330 sealed ACcoding rows, only the count and the first identifier have been read, and no result has been computed on the sealed parts of ACcoding and OULAD. Every access is recorded in the repository.

We opened the CodeBench terms once, on 25 September 2026, after the method, its parameters and the primary metric had been frozen, and Section 8.2 reports that evaluation. Every other number in this paper is a development number. A first attempt at the run stopped before producing any result, because a cleaning rule named only the development terms; the rule was extended to the sealed terms and the protocol frozen again (Supplementary Section S10).

A rule fixed before the grids were run chooses the guard's shape parameters. Each of the three shapes has a grid of parameter points, and the rule keeps the points whose worst harm over 15 validation cells is at most  $G/2$ . Among those it takes the one with the largest worst-cell gap closed (Supplementary Section S5). The validation cells are the five overlays at three loads, built on development terms other than the last.

However, the guard family and a coarse grid were explored on the evaluation overlays before this rule existed. The selection is therefore not held out from development, and we do not call any figure out of sample. Supplementary Section S5 records what the rule selected, which results the released code reproduces, and how the choice moves when one validation cell is dropped.

## 8. Experiments

Every policy is simulated on the shared pool of Section 7.2, with the same arrivals and costs, so each comparison is made job by job. Supplementary Table S4 lists the configuration.

### 8.1. The Headline: Scores That Use Only Available Results

The prediction study of Section 4 releases a job's result at the time the original platform recorded it. In the simulated queue, a job completes at a time that depends on the policy, so a score computed on the original clock may use a result that the simulated queue has not produced yet. The *online* protocol removes this for each copy of a class-term. It freezes the fitted model and simulates each policy once. When a job arrives, the model scores it from the results of earlier jobs in the same copy of its class-term that this simulation has completed by that moment, and from the recorded results of other classes and terms, which keep their original times. A score therefore follows the simulated queue for its job's own copy and the recorded clock for all others. Supplementary Section S3.7 defines the simulation and checks it against a second implementation.

Four references stand beside the online protocol. The *exact* protocol also freezes the model. It simulates a policy, withholds every result that was not yet complete when the job using it arrived, rescores, and simulates again until no score uses an unavailable result. A result it has withheld once stays withheld, even when a later simulation completes it in time, so it can use less history than the online protocol. The *original* protocol keeps the platform's clock and is optimistic. The *conservative* protocol admits a result only an hour after it appeared, uses only history from the same class-term copy, and refits the models.

The *static* protocol uses no outcome history at all. The guard's parameters are the same under all five.

At the busiest load, Guard(600) under the online protocol reaches a  $p99_{dl}$  of 111.25 s and closes 0.705 [0.648, 0.768] of the gap (Table 4). The same guard closes 0.672 [0.602, 0.743] under the exact protocol, 0.801 [0.764, 0.842] under the original one, 0.429 [0.357, 0.533] under the conservative one and 0.349 [0.281, 0.457] under the static one. The online figure lies between the exact and the original ones. The exact protocol uses less history than the online one, and the original protocol uses results the queue has not produced. At the two lighter loads the two figures are closer, 0.725 against 0.740 and 0.800 against 0.837, and their intervals overlap. Unguarded SPJF-E closes 0.864 under the online protocol. The conservative figure falls further than the exact one because it also withholds history that the queue would have had and refits the models. Supplementary Section S3.7 separates the two effects. The parameters were selected under the original protocol. The full grids were run under the original protocol, so the rest of this section uses it unless stated otherwise, and its figures are optimistic references.

**Table 4.** Guard(600) under five protocols for what the scores may use. Online: each score uses the results of its own class-term copy completed by its job's arrival in the simulated queue, and the recorded results of other classes and terms (the headline). Exact: results withheld and the queue simulated again until no score uses an unavailable result; a withheld result stays withheld. Original: the platform's own clock (optimistic). Conservative and static withhold more than the queue requires. Gap closed with 95% intervals from resampled weeks (Section 7.2), which do not cover the choice of classes and terms; waits, largest excess and harm in seconds. Fired: share of dispatch decisions, weighted by queue length, at which the guard overrode the base policy. At  $\rho = 0.5$ ,  $k = 7.8$  is the mean over the five overlays.

| Load                  | Protocol     | $p99_{dl}$ | Gap closed           | Max excess | Harm  | Fired (%) |
|-----------------------|--------------|------------|----------------------|------------|-------|-----------|
| $\rho = 0.5, k = 7.8$ |              |            |                      |            |       |           |
|                       | online       | 15.22      | 0.725 [0.696, 0.746] | 499.5      | 248.3 | 0.38      |
|                       | exact        | 15.25      | 0.724 [0.697, 0.747] | 498.6      | 248.1 | 0.45      |
|                       | original     | 14.97      | 0.740 [0.713, 0.760] | 492.2      | 237.0 | 0.17      |
|                       | conservative | 16.68      | 0.640 [0.604, 0.671] | 499.9      | 235.2 | 0.94      |
|                       | static       | 17.95      | 0.560 [0.519, 0.594] | 498.0      | 225.1 | 0.38      |
| $\rho = 0.8, k = 5$   |              |            |                      |            |       |           |
|                       | online       | 50.57      | 0.800 [0.772, 0.819] | 509.3      | 215.7 | 2.18      |
|                       | exact        | 51.90      | 0.785 [0.748, 0.807] | 509.0      | 215.6 | 2.64      |
|                       | original     | 47.47      | 0.837 [0.815, 0.851] | 507.8      | 319.6 | 1.31      |
|                       | conservative | 64.77      | 0.634 [0.560, 0.688] | 509.4      | 327.7 | 3.26      |
|                       | static       | 70.18      | 0.571 [0.504, 0.638] | 512.9      | 341.6 | 3.89      |
| $\rho = 1, k = 4$     |              |            |                      |            |       |           |
|                       | online       | 111.25     | 0.705 [0.648, 0.768] | 517.7      | 333.2 | 7.27      |
|                       | exact        | 118.94     | 0.672 [0.602, 0.743] | 519.9      | 332.8 | 7.82      |
|                       | original     | 89.24      | 0.801 [0.764, 0.842] | 523.7      | 339.9 | 7.15      |
|                       | conservative | 174.71     | 0.429 [0.357, 0.533] | 527.8      | 303.5 | 7.71      |
|                       | static       | 193.11     | 0.349 [0.281, 0.457] | 528.1      | 316.1 | 11.18     |

## 8.2. The Sealed Terms

The three sealed terms of the main log were opened once, on 25 September 2026, after the method, its parameters and the primary metric had been frozen (Section 7.3). The run replayed the frozen configuration and code and selected nothing, and the access ledger in the repository lists each of its ten commands and what each produced. The sealed terms hold fewer class-terms than the development pool, so each of their five overlays superposes 31 copies over 29 weeks. The number of servers follows the rule of Section 7.2, and the busiest hours reach utilisations of 0.487, 0.770 and 0.954 at the three loads.

At the busiest load, Guard(600) under the online protocol reaches a  $p99_{dl}$  of 47.22 s and closes 0.894 [0.862, 0.912] of the gap (Table 5). It closes 0.880 under the exact protocol and 0.911 under the original one, and the five protocols fall in the same order as on the development terms. Unguarded SPJF-E closes 0.936 [0.909, 0.948] online. The guard's largest excess over FCFS is 533.8 s against its promise of 600 s, and its harm is 237.5 s, where SPJF-E reaches 3,301.6 s and 1,602.0 s. No guard in the main comparison exceeds its promise at any load (Supplementary Section S10).

Every policy closes a larger share of the gap on the sealed terms than on the development ones, except five of the deliberately corrupted rankings, which do more damage there. The sealed run therefore confirms the guarantee and the order of the policies, but not the size of the development figures. The running-time predictor ranks the sealed terms as well as the development ones, with a pooled AUROC of 0.9414 for SPJF-E's score against 0.9360 on the development terms.

**Table 5.** Guard(600) under the five protocols of Table 4 on the three sealed terms, opened once after the method and its parameters were frozen. Columns as in Table 4;  $k$  is the mean over the five overlays.

| Load                  | Protocol     | $p99_{dl}$ | Gap closed           | Max exc. | Harm  | Fired (%) |
|-----------------------|--------------|------------|----------------------|----------|-------|-----------|
| $\rho = 0.5, k = 8.2$ |              |            |                      |          |       |           |
|                       | online       | 3.76       | 0.895 [0.872, 0.906] | 260.8    | 166.2 | 2.59      |
|                       | exact        | 3.77       | 0.893 [0.871, 0.903] | 256.6    | 166.6 | 3.29      |
|                       | original     | 3.75       | 0.897 [0.878, 0.907] | 264.0    | 173.9 | 4.01      |
|                       | conservative | 4.15       | 0.839 [0.816, 0.858] | 264.1    | 178.0 | 3.40      |
|                       | static       | 4.46       | 0.792 [0.744, 0.824] | 259.3    | 203.3 | 0.16      |
| $\rho = 0.8, k = 5.2$ |              |            |                      |          |       |           |
|                       | online       | 24.59      | 0.884 [0.862, 0.903] | 506.6    | 202.6 | 6.40      |
|                       | exact        | 24.74      | 0.880 [0.860, 0.899] | 499.2    | 202.6 | 4.41      |
|                       | original     | 24.22      | 0.891 [0.874, 0.908] | 504.6    | 189.5 | 7.01      |
|                       | conservative | 28.09      | 0.811 [0.772, 0.845] | 520.3    | 233.2 | 7.06      |
|                       | static       | 31.70      | 0.740 [0.691, 0.786] | 533.7    | 255.3 | 1.33      |
| $\rho = 1, k = 4.2$   |              |            |                      |          |       |           |
|                       | online       | 47.22      | 0.894 [0.862, 0.912] | 533.8    | 237.5 | 7.65      |
|                       | exact        | 49.40      | 0.880 [0.839, 0.903] | 526.5    | 243.1 | 5.25      |
|                       | original     | 44.33      | 0.911 [0.890, 0.926] | 522.9    | 293.2 | 7.10      |
|                       | conservative | 65.08      | 0.784 [0.712, 0.845] | 518.8    | 244.6 | 8.57      |
|                       | static       | 82.66      | 0.679 [0.583, 0.783] | 528.6    | 332.3 | 3.20      |

### 8.3. What to Rank By

Ranking by expected cost beats the log-scale score at the two higher loads and ties with it at the lightest. The paired difference in gap closed is  $-0.000 [-0.016, 0.013]$ ,  $+0.012 [0.004, 0.019]$  and  $+0.019 [0.011, 0.024]$  at  $\rho = 0.5, 0.8$  and  $1.0$ . Refitting the log-scale model from scratch moves this difference by at most 0.001 (Supplementary Table S16). Quantile scores and squared error on raw seconds lose to the Tweedie fit at every load. A rule that only separates predicted-heavy from predicted-light jobs recovers little of the benefit, so most of the benefit comes from the order inside the light class (Supplementary Table S14).

The two kinds of prediction error do not cost the same. Replacing the largest 1% of underestimates by the true cost closes 0.91 of the distance to SJF in  $p99_{dl}$  at  $\rho = 1.0$  on both overlays tested. Repairing the largest 1% of overestimates closes only 0.26 and 0.24. At 0.1%, the repair of underestimates closes 0.47 and that of overestimates 0.04. Treating a long job as short is thus the expensive mistake, by about an order of magnitude. The repair depends on the score's numerical scale, so this result describes our fitted predictor and not prediction errors in general.

#### 8.4. What the Guarantee Costs and Gains

Table 6 gives the guard at three bounds and three loads. At  $\rho = 1.0$ , Guard(600) closes 0.801 of the gap and the unguarded policy 0.916. A bound of ten time limits therefore costs 0.115 of the gap. The cost is 0.072 at  $G = 1200$  s and 0.438 at  $G = 300$  s. The paired intervals of these three differences are  $[-0.142, -0.077]$ ,  $[-0.083, -0.052]$  and  $[-0.526, -0.314]$ , and none contains zero, so the guarantee has a cost at every  $G$  we tested. What the guarantee buys is protection of the worst job. The unguarded policy delays one job at the busiest load by 5,904.5 s beyond its FCFS wait, 98.4 times the 60-second limit. The guard keeps every job within its bound. It also keeps the worst wait among heavy jobs within about 450 s of its FCFS value, whereas the unguarded policy multiplies that wait by 2.8 to 4.1.

**Table 6.** Guard( $G$ ) at three bounds and three loads, original protocol, with the FCFS and SPJF-E figures of each load in its header. Gap closed with 95% intervals. Reduction is the plain percentage reduction of  $p99_{dl}$  against FCFS, which is a different number. Largest excess and harm are the worst over the five overlays, and the other columns are means. Times in seconds. Fired as in Table 4. Supplementary Table S6 adds the comparators.

| Load                                                                                        | Policy      | G (s) | $p99_{dl}$ | Gap closed           | Reduction (%) | Max excess | Harm  | Fired (%) |
|---------------------------------------------------------------------------------------------|-------------|-------|------------|----------------------|---------------|------------|-------|-----------|
| $\rho = 0.5, k = 7.8$ ; FCFS $p99_{dl} = 27.64$ s, SPJF-E 14.68 s with max excess 1,241.4 s |             |       |            |                      |               |            |       |           |
|                                                                                             | Guard(300)  | 300   | 15.58      | 0.705 [0.678, 0.724] | 43.7          | 192.5      | 132.1 | 1.35      |
|                                                                                             | Guard(600)  | 600   | 14.97      | 0.740 [0.713, 0.760] | 45.8          | 492.2      | 237.0 | 0.17      |
|                                                                                             | Guard(1200) | 1200  | 14.78      | 0.750 [0.723, 0.772] | 46.4          | 996.7      | 106.0 | 0.09      |
| $\rho = 0.8, k = 5$ ; FCFS $p99_{dl} = 119.35$ s, SPJF-E 43.33 s with max excess 3,044.5 s  |             |       |            |                      |               |            |       |           |
|                                                                                             | Guard(300)  | 300   | 55.94      | 0.738 [0.697, 0.761] | 53.0          | 209.7      | 114.1 | 11.57     |
|                                                                                             | Guard(600)  | 600   | 47.47      | 0.837 [0.815, 0.851] | 60.1          | 507.8      | 319.6 | 1.31      |
|                                                                                             | Guard(1200) | 1200  | 46.31      | 0.850 [0.829, 0.865] | 61.1          | 1,096.1    | 117.7 | 0.41      |
| $\rho = 1.0, k = 4$ ; FCFS $p99_{dl} = 273.50$ s, SPJF-E 62.79 s with max excess 5,904.5 s  |             |       |            |                      |               |            |       |           |
|                                                                                             | Guard(300)  | 300   | 163.40     | 0.478 [0.387, 0.602] | 40.2          | 229.1      | 134.8 | 28.64     |
|                                                                                             | Guard(600)  | 600   | 89.24      | 0.801 [0.764, 0.842] | 67.3          | 523.7      | 339.9 | 7.15      |
|                                                                                             | Guard(1200) | 1200  | 79.32      | 0.844 [0.817, 0.870] | 70.9          | 1,110.4    | 123.4 | 1.40      |

The harm limit used in selection does not always carry over to the evaluation overlays. Guard(600) has harms of 319.6 and 339.9 s at the two higher loads, above the 300 s validation limit, although every excess stays within the bound. The shape of the budget matters mainly where the bound is tight and the load high. There it decides which jobs are delayed more than how much of the benefit survives (Supplementary Sections S3.3 and S3.11). Counting overtakes instead of work gains almost nothing on this load (Supplementary Figure S2). Overtakers here are mostly shorter than a second, while a count must be sized against the worst case of  $L$  per pass. Our count baseline is charged when an overtaker starts. Charging it at completion instead changes its gap closed by at most 0.015, and by at most 0.001 at the two higher loads. An aging baseline ranks by  $\hat{C}_i - \omega(t - a_i)$ , with the credit  $\omega$  chosen under the same harm limit. At the busiest load under the conservative protocol it closes 0.051 of the gap, against 0.429 for the guard, and one job exceeds its FCFS wait by 855.6 s. Aging carries no proved bound (Supplementary Table S12).

A maximum waiting time gives a guarantee of the same form (Proposition 9), so the guard's counter has to earn its place against it. Table 7 runs Timeout( $G - (3 - 2/k)L$ ) beside Guard( $G$ ), which gives both the bound  $G$ . At  $\rho = 0.5$  the two differ by at most 0.053 of the gap. Near saturation they part. At  $\rho = 1.0$  and  $G = 600$  s the clock closes 0.420 and the guard 0.801. At  $G = 300$  s the clock closes 0.031, and its  $p99_{dl}$  of 266.4 s is barely below FCFS's 273.5 s. At  $G = 1200$  s the two agree again. The clock's bound is not known to be tight, so we also compare the two at equal realised worst excess. On the first overlay at the busiest load, Timeout with  $\theta = 480$  s reaches a largest excess of 506.1 s and closes 0.490 of the gap. Guard(600) reaches 512.3 s and closes 0.797. The clock also harms more, 456.7 s against 339.9 s at  $\rho = 1.0$  and  $G = 600$  s.

**Table 7.** Guard(G) against the maximum waiting time  $\text{Timeout}(G - (3 - 2/k)L)$ , which has the same bound G. Gap closed, mean over the five overlays, with 95% intervals. The difference is Guard minus Timeout, resampled as a difference. Harm is the worst over the overlays, in seconds. Original protocol.

| Load | G (s) | Guard(G)             | Timeout              | Difference              | Harm, Guard / Timeout |
|------|-------|----------------------|----------------------|-------------------------|-----------------------|
| 0.5  | 300   | 0.705 [0.678, 0.724] | 0.652 [0.619, 0.679] | +0.053 [+0.030, +0.077] | 132.1 / 148.0         |
|      | 600   | 0.740 [0.713, 0.760] | 0.734 [0.707, 0.753] | +0.006 [+0.001, +0.012] | 237.0 / 237.0         |
|      | 1200  | 0.750 [0.723, 0.772] | 0.752 [0.724, 0.774] | −0.002 [−0.005, +0.001] | 106.0 / 237.0         |
| 0.8  | 300   | 0.738 [0.697, 0.761] | 0.244 [0.017, 0.508] | +0.494 [+0.241, +0.691] | 114.1 / 177.1         |
|      | 600   | 0.837 [0.815, 0.851] | 0.789 [0.754, 0.807] | +0.048 [+0.030, +0.080] | 319.6 / 443.7         |
|      | 1200  | 0.850 [0.829, 0.865] | 0.857 [0.837, 0.869] | −0.006 [−0.012, −0.001] | 117.7 / 469.2         |
| 1.0  | 300   | 0.478 [0.387, 0.602] | 0.031 [0.012, 0.055] | +0.447 [+0.369, +0.550] | 134.8 / 182.3         |
|      | 600   | 0.801 [0.764, 0.842] | 0.420 [0.131, 0.673] | +0.381 [+0.158, +0.650] | 339.9 / 456.7         |
|      | 1200  | 0.844 [0.817, 0.870] | 0.827 [0.786, 0.865] | +0.017 [−0.002, +0.044] | 123.4 / 708.8         |

Ordering by predicted cost inside fixed windows, as in the closest scheduler for judging queues [53], limits overtaking by position. We ran a version of it on the first overlay with windows of up to 512 jobs. It closes at most 0.417, 0.146 and 0.057 of the gap at the three loads, with largest excesses of 232.7, 359.6 and 444.5 s. On the same overlay Guard(300) closes 0.678, 0.716 and 0.486, with 189.3, 199.7 and 219.6 s. Near saturation, a window large enough to help lets long jobs wait longer than the guard does. The last test corrupts the scores deliberately, by reversing them, by drawing them at random, or by calling the longest 1% of jobs the shortest. The unguarded policy then delays some job by more than 6,000 s beyond FCFS at the busiest load, and Guard(600) keeps every job within its bound. The guarded  $p_{99_{dl}}$  is then 546.5, 261.9 and 543.6 s, against FCFS's 273.50 s. The guarantee thus protects the worst single job but not the shape of the tail, and a guarded run may have a worse percentile than FCFS, by at most G (Supplementary Table S8).

#### 8.5. The Bounds on the Traces

Every run that includes a bounded policy checks the bound on every job inside the simulation, so a violation would expose an error in the code rather than in the theorem. The 315 policy-by-cell runs reported here make 5,554,949,400 such checks. The single-server runs add 67,396,287, and an independently written second simulator adds 5.7 billion. None of them finds a violation, and the count of violations is 0. Proposition 9 was checked job by job on the same overlays and Proposition 10 on all 304 bounded runs, also with 0 violations. Proposition 11 was checked on constructed instances (Supplementary Section S4.12). The ratio of the realised worst excess to the bound falls as  $k$  grows, from 0.9950 at  $k = 1$  to 0.8850 at  $k = 8$ . At one server Guard(600) reaches 595.5 s of its 600 s bound (Supplementary Section S3.6). The residual  $\Delta_i$  of Theorem 2 never exceeds 0.518 of its own bound  $2(k - 1)L$  (Supplementary Sections S3.13 and S3.16).

The absolute bound of Proposition 10 comes close to the realised waits. At one server every FCFS wait equals its fluid backlog to the microsecond, as the proposition predicts, and the largest wait reaches 0.999 of the bound. At the three loads the worst wait under a work budget reaches 0.886, 0.951 and 0.963 of the bound, and under FCFS 0.881, 0.945 and 0.965. An absolute bound quoted in advance needs a forecast of the burst  $\sigma$ . The burst of the validation terms predicts that of the evaluation terms only to within a factor of 0.738 to 1.573, and it is too low in 10 of 15 cells. A bound quoted from it is broken in 18 of 60 runs, by up to 349.8 s. Such a bound therefore needs a margin, here a factor of 1.36 on  $\sigma$ , or an admission control that keeps the arrivals within the quoted burst.

### 8.6. Where the Method Applies

Table 8 places the main log beside the other workloads. Excesses are in units of each log's  $L$ , the unit in which the guarantee is written. For the grading logs  $L$  is the time limit, and for the others it is the analysis cap of Section 7.1. Two screens say where the guard is useful. The first compares the FCFS tail with the guarantee's additive term  $(3 - 2/k)L$ , which is close to  $3L$ . If the FCFS p99 is below this term, the guarantee is coarser than the waits themselves. CodeBench, at  $4.56L$ , and the cloud-function log, at  $4.22L$ , clear the first screen, and four of the six workloads do not. The compute farm is the extreme case, with an FCFS p99 of 18.7 h against  $L = 493,857$  s, or 5.7 days. The second screen measures how concentrated the costs are, which is the opportunity for reordering. In the two CI pools the longest 1% of jobs carry only 3.4% and 5.2% of the work. There, ordering by true cost makes the p99 worse in three of the four windows we examined. Where both screens are cleared, the guard closes 0.48 to 0.71 of the gap and holds the largest excess to 2.5 to  $3.8L$  at a bound of  $5L$ . Clearing both screens does not guarantee a gain, as the next subsection shows.

**Table 8.** The workloads under one format. Tail share: fraction of executed work carried by the longest 1% of jobs. AUROC: heavy label from information available at arrival. Screen 1: whether the FCFS p99 exceeds  $(3 - 2/k)L$ . Excess in units of each log's  $L$ . A negative gap closed means that the policy's p99 is worse than FCFS's. CodeBench row: original protocol. Supplementary Section S6.7 describes the CI capacities and adds a log from a grid computing cluster.

| Log and load                             | Tail share | AUROC | FCFS p99/ $L$ | Screen 1 | Gap closed, SPJF-E | Unguarded max excess | Guard at $G \approx 5L$ gap closed / max excess |
|------------------------------------------|------------|-------|---------------|----------|--------------------|----------------------|-------------------------------------------------|
| CodeBench, $\rho \approx 1.0$            | 0.375      | 0.936 | 4.56          | yes      | 0.916              | 98.4L                | 0.478 / 3.82L                                   |
| ACcoding, drawn arrivals, $\rho = 0.8^a$ | 0.122      | 0.927 | 0.021         | no       | 0.980              | 20L                  | 0.974 / 2.21L                                   |
| Azure Functions, $\rho = 1.0$            | 0.443      | 0.985 | 4.22          | yes      | 0.858              | 10.07L               | 0.710 / 2.52L                                   |
| CI pool A, $k = 156$ of 173              | 0.034      | 0.651 | 1.78          | no       | n/a <sup>b</sup>   | 11.53L               | 0.735 (mean) <sup>b</sup> / 2.64L               |
| CI pool B, $k = 96$ of 105               | 0.052      | 0.997 | 0.373         | no       | 0.958              | 0.34L                | 0.958 / 0.34L                                   |
| Intel Netbatch, $\rho = 0.95$            | 0.372      | 0.890 | 0.136         | no       | -4.111             | 0.85L                | -4.346 / 0.83L                                  |

<sup>a</sup> ACcoding arrivals are drawn, so this row describes the ordering and not waiting on that platform. Its budget is the largest tested,  $B = 10L$ , and its tail share is simulated with  $L = 30.5$  s (0.275 on measured cost at  $L = 30.0$  s, the figure of Section 1). <sup>b</sup> On CI pool A, SJF is itself worse than FCFS at the p99, so no p99 gap is defined; the guard's entry is the gap in mean wait and is not comparable with the other rows.

### 8.7. Capacity, Physical Runs and Feedback

The benefit depends on capacity. We ran every  $k$  from 1 to 20 on the five overlays. At  $k = 1$ , Guard(600) makes p99<sub>dl</sub> worse by 493.79 s, and all five overlays agree that it improves the p99 only for  $k = 4$  to 12. We also ran the guard physically, on two worker processes that occupy real time slots for each job's recorded duration. This run executed 2,151 jobs of one development window, and all 6,453 dispatch decisions followed the rule. The guarded p99 of 1,783.65 s was nevertheless no better than FCFS's 1,763.30 s, and single jobs differed from the ideal simulation by up to 1,354.10 s. A second physical test used the two-worker back end of a course platform, where twenty scripted accounts submitted 100 jobs per cell in a deadline-shaped burst with  $G = 400$  s. Unguarded predicted order left 7 jobs beyond the bound. The four guarded cells left none, with a worst excess of 0.514G (Supplementary Sections S6.1, S6.2 and S6.9).

The simulations replay arrivals at their recorded times, so they neither reward a scheduler with the work it would attract nor punish it for the users it would drive away [78,79]. The logs record one part of that feedback, the time between one result and the same student's next submission. In a second replay, each submission is released only after its predecessor's result plus that recorded delay. This replay shortens p99<sub>dl</sub> by a factor of 1.3 to 3.8. Guard(600) then closes 0.825, 0.809 and 0.810 of the gap at the three loads,

against 0.734, 0.827 and 0.800 with the recorded arrival times, and it keeps every job within its bound (Supplementary Section S6.5).

## 9. Limitations

**The queue is constructed, and the replay is open-loop.** CodeBench gives each student a separate container [77], so its logs do not record waiting in a shared queue [21]. We use these logs as a record of demand, with reconstructed arrival times and recorded running times. The pool, the queue and the scheduler are ours, and nothing here describes waiting on that platform. A fixed pool of grading machines behind one queue is a standard deployment [7,8], and on the same demand 14 servers replace 1,636 concurrent containers [80] (Supplementary Section S2.3). Arrivals are replayed at their recorded times, so a scheduler is neither rewarded with the work it would attract nor punished for the users it would drive away. Replaying each student's recorded delay between one result and the next submission (Section 8.7) restores one part of that feedback and does not reverse the benefit. It restores neither abandonment nor changes in what students submit, however, and the recorded delays come from a platform on which no student waited. Our results are therefore conditional on a fixed amount of offered work, and Supplementary Section S6.3 constructs feedback under which a scheduling advantage disappears or reverses. We raise the load by overlaying whole class-terms aligned on weekday and clock time, rather than by compressing the gaps between arrivals [81], which would also compress the daily cycle [78].

**The guarantee is written in units of  $L$ , and per job.** Every additive constant in Section 6 is a multiple of the time limit, apart from the report delay  $\delta$  and the arrival burst  $\sigma$ , and the guard's constant  $(3 - 2/k)L$  cannot be improved. No wrapper that reorders at all can guarantee less than  $L$  around every base policy (Proposition 12). Stopping long jobs and restarting them in a second tier, as task assignment by guessing size does [82], does not escape this floor on a shared pool. The stopped work is lost rather than overtaken, and the second tier is again non-preemptive in blocks of  $L$  (Supplementary Section S7.5). Charging a cap known at arrival lowers the constant by exactly  $L$  and leaves the floor. Because the guarantee bounds each job, it also bounds each percentile,  $Q_\alpha(W_P) \leq Q_\alpha(W_{\text{FCFS}}) + G$ . It does not ensure that a percentile improves, and under corrupted scores a guarded run had a worse  $p_{99_{\text{dl}}}$  than FCFS (Section 8.4). Where the FCFS tail is already below about  $3L$ , the bound is coarser than the waits it bounds, which is the case for four of the six workloads (Section 8.6).

**Development results and the sealed terms.** We explored the guard family and an early parameter grid on the overlay the tables report, and only then searched the pre-stated grids on validation overlays that exclude the evaluation term. The selection is thus validation-only within a family that is not. The worst-cell rule is sensitive to which cells it sees, and the five overlays are week-shifts of the same class-terms rather than independent samples. Resampling whole class-terms was run under the original protocol only, so the online figures carry intervals from resampled weeks, which do not cover the choice of classes and terms. Moreover, the shape selected on 15 cells with  $k \geq 4$  does not carry over to a single server (Supplementary Section S3.14). We therefore do not call the parameters validated. The guarantee does not depend on them, and only the benefit does. The sealed terms were opened once, after the method, parameters and metric were frozen (Section 8.2). They come from the same platform and pass through the same constructed queue, so they test the frozen choices on new classes and terms but not on another service.

**Information, costs, and what was not studied.** Only the cells of the main comparison were replayed under the online protocol, and the guard's parameters were selected under the original one, so the comparisons of Sections 8.3 to 8.7 rest on the optimistic

original protocol. The online protocol takes the results of other classes and terms at their recorded times. The recorded costs are wall-clock durations that already contain some contention, and the 60-second limit is inferred from the data rather than documented. AC-coding arrival times are drawn because the released tables omit them (Supplementary Section S2.1). Preemption is outside the model. The wrapper applies unchanged to a learned or reinforcement-learned base policy, which we did not test. Capacity is fixed within each replay, and the capacity sweep is a sensitivity, not a planning study. Computing the features and running the predictor are charged only in a sensitivity run, where they move the gap closed by less than 0.001 (Section 4). We ran a physical service but no production deployment, and we measured no user response. Nor did we combine the guard with an existing scheduler such as the serverless prototype of Fu et al. [59], which has no proved per-job bound.

## 10. Conclusions

A service that runs many users' jobs on a few shared servers can order its queue by predicted running time and still guarantee every job a wait no more than  $G$  seconds longer than in arrival order, whatever the predictions are. This guarantee follows from one identity. A job's extra wait relative to FCFS equals the net running time of the jobs that overtook it, divided by the number of servers, up to at most  $(2 - 2/k)L$ . Limiting that overtaken work is therefore sufficient for a per-job guarantee and, up to an additive  $O(kL)$ , necessary. The guard enforces such a limit with the exact constant  $(3 - 2/k)L$ , and its guarantee survives completion reports that arrive up to  $\delta$  late at a cost of  $L + \delta$ . A maximum waiting time gives a guarantee of the same form. Near saturation and at tight bounds, however, it keeps much less of the benefit, because it charges a job for all the work ahead of it and not only for the jobs that pass it. Counting overtakes gives the same guarantee only at a charge of  $L$  per pass, which on the main log is 150 times the mean running time.

For the grading system that opened this paper, the answer to a student whose submission sat behind a looping one is now a number. At the busiest load of the recorded course demand, with every score computed from the results available when its job arrives, a guard with  $G = 600$  s keeps 0.705 of the reduction in 99th-percentile wait before deadlines that SJF achieves over FCFS. Unguarded ordering keeps 0.864, and with the guard no job exceeds its bound. On three terms held back until the method was frozen, the same guard keeps 0.894 and unguarded ordering 0.936. The scores come from gradient-boosted trees on each student's and each task's history, chosen on queueing outcomes rather than prediction accuracy. A heterogeneous graph network over students, tasks, assignments and cohorts does not beat these trees on average. Its edges add 0.006 in rank correlation over all 40,844 test submissions and 0.028 on the 1,092 submissions to tasks with fewer than five earlier results, the situation that a newly released assignment creates. The demand is constructed in both evaluations, and the benefit depends on capacity: it appeared only for  $k = 4$  to 12.

Two questions remain open. Under the executed-work convention of Section 6.1, the lower constant of the identity is known only to lie in  $[(k - 1)L, 2(k - 1)L]$ . We also do not know whether some wrapper overrides the base policy strictly less often than Algorithm 1 while guaranteeing the same  $G$ .

**Supplementary Materials:** The following supporting information can be downloaded at <https://www.mdpi.com/article/10.3390/math1010000/s1>: Table S1: Predictors on the non-judging logs, by feature group; Table S2: Predictors on the two grading logs under one protocol; Table S3: Share of work carried by the longest jobs, and the provenance of  $L$ ; Table S4: Experimental configuration and selection protocol; Table S5: Selected budget points and per-shape grid winners; Table S6: The

guard at three bounds, with the comparators; Table S7: The age-relative and queue-length winners; Table S8: Deliberately corrupted scores; Table S9: The residual of Theorem 2 on nine cells; Table S10: The single-server configuration; Table S11: Definition of linear aging; Table S12: Comparison with the aging baseline; Table S13: Predictor AUROC with user-block intervals; Table S14: Alternative ranking scores; Table S15: Trace-construction sensitivities; Table S16: Unguarded policies; Table S17: Candidates and feasible points of the selection; Table S18: Capacity sweep; Table S19: Physical timed-work runs; Table S20: Replay with recorded think times; Table S21: A grid log by walltime class; Table S22: The grid log's 900-second class at its own limit; Table S23: The grid-log rows of the cross-log comparison; Table S24: Six cells on a course platform's back end; Table S25: Scope of the nearest per-job guarantees; Figure S1: Gap-harm trade-offs of the budget shapes; Figure S2: What each budget shape keeps as the bound is relaxed; Figure S3: Original-score reference and corrupted-score controls. The Supplementary Materials also give the proofs deferred from Section 6 (Sections S4, S7 and S8) and extended related work (Section S9).

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## Abbreviations

The following abbreviations are used in this manuscript:

|          |                                                         |
|----------|---------------------------------------------------------|
| AUROC    | Area under the receiver operating characteristic curve  |
| CI       | Continuous integration                                  |
| EASY     | Extensible Argonne Scheduling sYstem                    |
| FCFS     | First-come first-served                                 |
| GNN      | Graph neural network                                    |
| GRU      | Gated recurrent unit                                    |
| OULAD    | Open University Learning Analytics Dataset              |
| p99      | 99th percentile                                         |
| R-GCN    | Relational graph convolutional network                  |
| RMSE     | Root mean squared error                                 |
| SJF      | Shortest job first                                      |
| SOAP     | Schedule Ordered by Age-based Priority                  |
| SPJF-E   | Shortest predicted job first on the expected-cost score |
| SPJF-log | Shortest predicted job first on the log-scale score     |
| SRPT     | Shortest remaining processing time                      |

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